Experiment 1: Word and Sentence Tokenization

1(a) Tokenize a sentence into words

```
import nltk

# Download tokenizer resources
nltk.download('punkt')
nltk.download('punkt_tab')

# Input sentence from the lab manual
text = (
    "This is a Demo Text for NLP using NLTK. "
    "Full form of NLTK is Natural Language Toolkit"
)

# Tokenize the sentence
word_tokens = nltk.word_tokenize(text)

# Display the tokens
print("Original Text:", text)
print("Word Tokens:", word_tokens)
```

```
Original Text: This is a Demo Text for NLP using NLTK. Full form of NLTK is N
Word Tokens: ['This', 'is', 'a', 'Demo', 'Text', 'for', 'NLP', 'using', 'NLTK
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

1(b) Split a paragraph into sentences

```
import nltk
from nltk.tokenize import sent_tokenize

nltk.download('punkt')
nltk.download('punkt_tab')

# Accept a paragraph
text = input("Enter a paragraph:\n")

# Split the paragraph into sentences
sentences = sent_tokenize(text)

# Display numbered sentences
for i, sentence in enumerate(sentences, start=1):
    print(f"Sentence {i}: {sentence}")
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
Enter a paragraph:
Hello there! Welcome to the NLP course. We will learn how machines understanc
Sentence 1: Hello there!
Sentence 2: Welcome to the NLP course.
Sentence 3: We will learn how machines understand language.
```

Experiment 2: Text Normalization

```
!pip -q install inflect spacy

import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
True
```

```
import string

text = "Hello! How are you doing today? I'm excited about NLP :) #AI"

clean_text = text.translate(
    str.maketrans('', '', string.punctuation)
)

print("Original:", text)
print("Without Punctuation:", clean_text)
```

```
Original: Hello! How are you doing today? I'm excited about NLP :) #AI
Without Punctuation: Hello How are you doing today Im excited about NLP  AI
```

```
import string
from nltk.tokenize import word_tokenize
```

```
text = "Hello! How are you doing today? I'm learning NLP, it's amazing :)"

tokens = word_tokenize(text)

# Remove individual punctuation tokens
clean_tokens = [
    word for word in tokens
    if all(char in string.punctuation for char in word)
    is False
]

clean_text = ' '.join(clean_tokens)

print("Original Text:", text)
print("Text without punctuation:", clean_text)
```

```
Original Text: Hello! How are you doing today? I'm learning NLP, it's amazing
Text without punctuation: Hello How are you doing today I 'm learning NLP it
```

```
text = (
    "This is a Demo Text for NLP using NLTK. "
    "Full form of NLTK is Natural Language Toolkit"
)

print("Lowercase:", text.lower())
print("Uppercase:", text.upper())
```

```
Lowercase: this is a demo text for nlp using nltk. full form of nltk is natur
Uppercase: THIS IS A DEMO TEXT FOR NLP USING NLTK. FULL FORM OF NLTK IS NATUR
```

```
from nltk.tokenize import word_tokenize

abbrev_dict = {
    "asap": "as soon as possible",
    "fyi": "for your information",
    "idk": "i don't know"
}

text = "FYI, I need it ASAP. IDK what happened."
tokens = word_tokenize(text)

expanded_tokens = []

for token in tokens:
    replacement = abbrev_dict.get(token.lower(), token)
    expanded_tokens.extend(replacement.split())

expanded_text = ' '.join(expanded_tokens)
```

```
print("Original:", text)
print("Expanded:", expanded_text)
```

```
Original: FYI, I need it ASAP. IDK what happened.
Expanded: for your information , I need it as soon as possible . i don't know
```

```
import inflect
from nltk.tokenize import word_tokenize

p = inflect.engine()

text = "I have 2 apples and 30 bananas."
tokens = word_tokenize(text)

converted_tokens = [
    p.number_to_words(word) if word.isdigit() else word
    for word in tokens
]

converted_text = ' '.join(converted_tokens)

print("Original:", text)
print("Converted:", converted_text)
```

```
Original: I have 2 apples and 30 bananas.
Converted: I have two apples and thirty bananas .
```

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

stop_words = set(stopwords.words('english'))

text = (
    "This is a Demo Text for NLP using NLTK. "
    "Full form of NLTK is Natural Language Toolkit"
)

tokens = word_tokenize(text)

# Case-insensitive stop-word removal
filtered_tokens = [
    word for word in tokens
    if word.lower() not in stop_words
]

print("Tokens after stop-word removal:", filtered_tokens)
```

```
Tokens after stop-word removal: ['Demo', 'Text', 'NLP', 'using', 'NLTK', '.',
```

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer

nltk.download('punkt')
nltk.download('punkt_tab')

stemmer = PorterStemmer()

text = (
    "It originated from the idea that there are readers who "
    "prefer learning new skills from the comforts of their drawing rooms"
)

tokens = word_tokenize(text)

for word in tokens:
    print(f"Actual: {word:12s} Stem: {stemmer.stem(word)}")
```

```
Actual: It           Stem: it
Actual: originated   Stem: origin
Actual: from         Stem: from
Actual: the          Stem: the
Actual: idea         Stem: idea
Actual: that         Stem: that
Actual: there        Stem: there
Actual: are          Stem: are
Actual: readers      Stem: reader
Actual: who          Stem: who
Actual: prefer       Stem: prefer
Actual: learning     Stem: learn
Actual: new          Stem: new
Actual: skills       Stem: skill
Actual: from         Stem: from
Actual: the          Stem: the
Actual: comforts     Stem: comfort
Actual: of           Stem: of
Actual: their        Stem: their
Actual: drawing      Stem: draw
Actual: rooms        Stem: room
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

```
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize

lemmatizer = WordNetLemmatizer()

text = "The striped bats were hanging on their feet and eating better mice"
tokens = word_tokenize(text)
```

```
for word in tokens:
    noun_lemma = lemmatizer.lemmatize(word, pos='n')
    verb_lemma = lemmatizer.lemmatize(word, pos='v')

    print(
        f"Actual: {word:12s} "
        f"Noun lemma: {noun_lemma:10s} "
        f"Verb lemma: {verb_lemma}"
    )
```

```
Actual: The          Noun lemma: The        Verb lemma: The
Actual: striped      Noun lemma: striped    Verb lemma: strip
Actual: bats         Noun lemma: bat        Verb lemma: bat
Actual: were         Noun lemma: were       Verb lemma: be
Actual: hanging      Noun lemma: hanging    Verb lemma: hang
Actual: on           Noun lemma: on         Verb lemma: on
Actual: their        Noun lemma: their      Verb lemma: their
Actual: feet         Noun lemma: foot       Verb lemma: feet
Actual: and          Noun lemma: and        Verb lemma: and
Actual: eating       Noun lemma: eating     Verb lemma: eat
Actual: better       Noun lemma: better     Verb lemma: better
Actual: mice         Noun lemma: mouse      Verb lemma: mice
```

```
!python -m spacy download en_core_web_sm -q
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 41.6 MB/s eta 0:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting th
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy

nlp = spacy.load("en_core_web_sm")

text = "The striped bats were hanging on their feet and eating better
doc = nlp(text)

for token in doc:
    print(f"Actual: {token.text:12s} Lemma: {token.lemma_}")
```

```
Actual: The          Lemma: the
Actual: striped      Lemma: striped
Actual: bats         Lemma: bat
Actual: were         Lemma: be
Actual: hanging      Lemma: hang
Actual: on           Lemma: on
Actual: their        Lemma: their
Actual: feet         Lemma: foot
```

```
Actual: and          Lemma: and
Actual: eating       Lemma: eat
Actual: better       Lemma: well
Actual: mice         Lemma: mouse
```